InsightfulCell Product Report

A one-stop solution for trustworthy, private data analysis

Prepared for
CDU IT Code Fair Sydney 2026 · Category: Coding Competition

Prepared by
Rupesh Ghimire (s403354), Product Owner and Lead Engineer
Sandesh Prasad Paudel (s388230), Quality Assurance and Deployment Engineer
Master of Data Science, Charles Darwin University

Date: 25 October 2026

Live application: https://app.ic.rgproject.me
Source code: https://github.com/rupeshghimire7/InsightfullCell
Demo video: [YouTube link]
Contact: rupeshghimire17@gmail.com

Contents

Executive summary....4
1 Introduction and background....5
1.1 Background....5
1.2 Problem statement....5
1.3 The idea....5
1.4 Objectives....6
1.5 Project plan and execution....6
1.6 Scope of this report....7
2 The product: InsightfulCell....8
2.1 Product at a glance....8
2.2 Upload any spreadsheet....8
2.3 Automatic understanding and privacy protection....9
2.4 Cleaning you can review....10
2.5 Statistically tested findings....10
2.6 A report where every number shows its work....13
2.7 Dashboards, maps and exploration....13
2.8 Download and share....14
2.9 Works with or without AI....15
2.10 Accessible by design....15
2.11 Handling everyday situations....16
2.12 Feature summary....17
3 Methodology: technical implementation....18
3.1 Development approach....18
3.2 Team roles and task division....18
3.3 System architecture....19
3.4 Technology stack....20
3.5 How the analysis works....20
3.6 Software engineering principles....23
3.7 Security and privacy by design....23
3.8 Testing and quality assurance....24

3.9 Code quality and documentation....25
3.10 Deployment and operations....25
4 Discussion....27
4.1 Market need and opportunity....27
4.2 Target customers and use cases....27
4.3 Competitive positioning....27
4.4 Business model and cost to serve....28
4.5 Scalability and go-to-market....28
4.6 Ethical and social impact....29
4.7 Current limitations....29
4.8 Future improvements and roadmap....29
4.9 Risks and mitigations....30
4.10 SWOT summary....30
5 Conclusion....31
Appendices....32
Appendix A. Evidence charts....32
Appendix B. Delivery timeline....33
Appendix C. Additional screenshots....34
Appendix D. Sample output....35
Appendix E. Technology detail....36
Appendix F. Glossary....36
Appendix G. How to evaluate InsightfulCell....37
Appendix H. Acknowledgements and attribution....37
References....39

Executive summary

InsightfulCell is a one-stop solution for data analysis: upload a spreadsheet and receive a cleaned dataset, statistically tested findings, checked charts and maps, and a plain-language report in which every number traces back to the test that produced it.

Organisations of every size now hold more data than they have people to analyse it. They face a trust gap. General AI assistants are easy to use, but they can invent numbers, hide how they cleaned the data and need the raw rows. Business intelligence and statistics tools are accurate, but they need a trained analyst to prepare the data and choose the tests.

InsightfulCell closes that gap with a strict division of labour. Code computes every number: parsing, cleaning, statistical tests, chart values and map values. An AI model, when one is switched on, only plans the questions and phrases the sentences, and everything it writes is verified against stored facts before anyone sees it. Raw rows never reach an AI model, and the product works fully without one.

Key results

Result	Evidence
14 seconds from upload to report	5,000 rows × 19 columns on the 1 GB live server, 22 October 2026
0 of 22 cross-tenant attacks succeeded	99-check security test, 18 October 2026
31 of 31 planted errors rejected	Sentence verifier evaluation, run with every test suite
1,439 chart and map values recomputed, 0 withheld	Zomato case study, 25 October 2026
1,275 automated tests; 93.04% backend coverage	Quality gate, 24 October 2026
US$0 model cost on the live site	Rules plan the analysis and the engine writes the report

InsightfulCell serves small and medium businesses, community organisations, students, researchers and consultants who need defensible answers without an analyst. It runs on a single free-tier cloud server for about US$3.60 a month (estimated), and with an AI model switched on, a measured analysis costs about US$0.02 in model fees.

Next, users will ask questions of their data in plain language and the service gains staging, monitoring and rate limits (Release 1 alpha, 11 December 2026), followed by team workspaces and shared reports (Release 2 beta, 12 March 2027).

1 Introduction and background

1.1 Background

Australia had 2,814,778 actively trading businesses at 30 June 2026, and 996,203 of them employed staff (Australian Bureau of Statistics [ABS], 2026). Most are small. They keep sales, bookings, survey and operations data in spreadsheets, yet few employ anyone whose job is statistics. Community organisations, public-sector teams and students are in the same position: the data exists, the analyst does not.

Generative AI made analysis feel easy, but fluent text is not the same as correct numbers. Language models are known to produce confident statements that their inputs do not support, a behaviour the research literature calls hallucination (Ji et al., 2023). For a business decision, an invented average is worse than no answer.

1.2 Problem statement

People who own data choose today between two imperfect options, compared in Table 1.

Table 1. The trust gap between the two common options

	General AI assistants	BI and statistics tools
Ease of use	High: ask in plain words	Low: needs a trained analyst
Numbers	Can be invented, with no way to check where they came from	Correct when used correctly
Data cleaning	Invisible	Manual, and easy to get wrong
Raw data	Sent to the AI provider	Stays local
Many tests at once	Rarely corrected for chance findings	Depends on the analyst

The gap is trust: the easy option cannot be trusted, and the trustworthy option is not easy.

1.3 The idea

We started from one question: *can analysing your own data be as easy as asking an assistant, and as trustworthy as the work of a careful analyst?* We turned it into four principles, in priority order, so that a conflict between them always has an answer (Table 2).

Table 2. The four principles and what each one required

Priority	Principle	What it required us to build
1	Correctness over fluency	Sentences whose numbers are filled in by code; a seven-check sentence verifier; charts withheld on any mismatch
2	Traceability	Unchangeable, hashed records of every finding; a source chip on every number; a reproducible cleaning recipe

Priority	Principle	What it required us to build
3	Privacy	Personal data masked the moment a file is read; no rows ever sent to an AI model; isolation enforced by the database
4	Simplicity	One codebase; one database that also runs the job queue; a complete product with no AI model at all

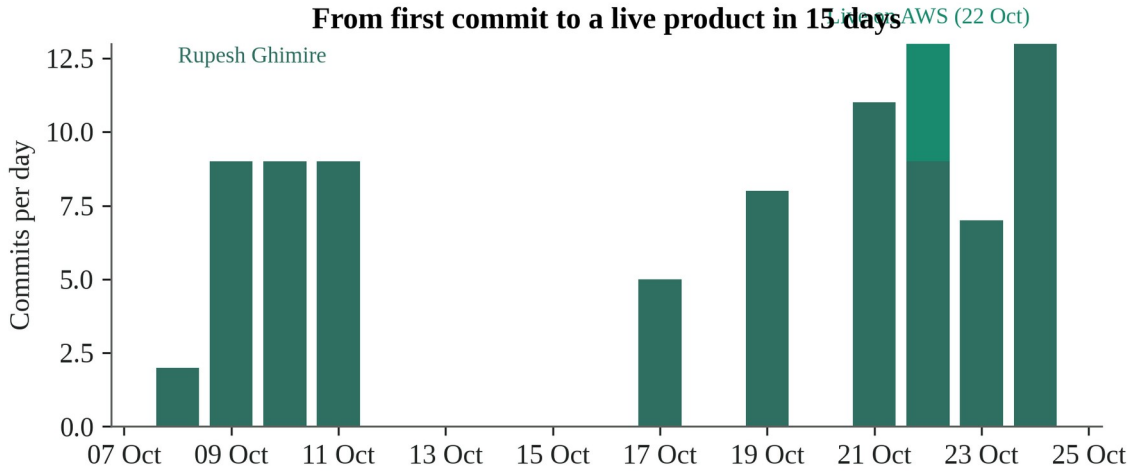
1.4 Objectives

Table 3. Project objectives and their status

Objective	Measure	Status	Evidence
Turn a spreadsheet into a checked report without an analyst	Upload to report in under a minute for a typical file	Achieved	14 s for 5,000 × 19 on the live server; 25 s for the 9,551-row case study, including the review step
Make every number traceable	Every number in the report links to a stored test	Achieved	Source chips on every number; 26 facts in the case study
Control false findings	False alarms limited across each analysis	Achieved	5% Benjamini–Hochberg control; 0 reliable findings on a pure-noise dataset
Keep raw data away from AI	No row or personal value reaches a model	Achieved	Masked description only; payload block; demo runs with no model
Run reliably in public	Live, secure, tested service	Achieved	Live on AWS since 22 October 2026; 0 of 22 cross-tenant attacks
Answer questions in plain language	Questions answered with the same checks	In progress	Planned for Release 1 alpha (11 December 2026)

1.5 Project plan and execution

We wrote the requirements and the system design on 7 October 2026, then built in staged releases, each closed only when every automated quality gate passed. Figure 1 shows the 15 days from the first commit to a live product: sign-in and uploads (8–9 October), the job queue and the first analysis steps (10 October), testing, verified writing, checked charts and exports (11 October), review modes and deletion (16–17 October), a deep system test (18 October), cost controls (21 October) and the public launch on Amazon Web Services (22 October).



Source: git log of main, 8–24 Oct 2026 (86 commits)

Figure 1. Commits per day from the first commit to the live product

1.6 Scope of this report

This report covers InsightfulCell as built and live on 25 October 2026. Features that are planned but not built are marked with their target release. Section 2 follows a user through the product, Section 3 explains how we engineered it, and Section 4 discusses its value, limitations and future.

2 The product: InsightfulCell

2.1 Product at a glance

InsightfulCell takes a file from upload to a shareable, checked report in six stages: upload, understand, clean, analyse, explain, and explore and share. Throughout this section we follow one real case study: the public Zomato restaurants dataset (9,551 restaurants in 15 countries, 21 columns, released into the public domain on Kaggle), analysed in Manual mode on 25 October 2026. Every screenshot and number below comes from that analysis.

Table 4. The six stages a user moves through

Stage	What the user does	What InsightfulCell does
Upload	Drops a file	Stores it securely, detects duplicates
Understand	Nothing	Identifies each column’s meaning; masks personal data
Clean	Reviews proposed changes (Manual mode)	Proposes fixes; applies approved ones to a copy
Analyse	Nothing	Tests every planned question with false-alarm control
Explain	Reads the report	Writes verified sentences; checks every chart value
Explore and share	Checks rows, explores a group, downloads	Opens the data, runs follow-ups, exports four formats

2.2 Upload any spreadsheet

What it does: accepts CSV, TSV, Excel, Parquet and JSON files up to 250 MB and sends them straight from the browser to secure storage in 8 MB parts.

Why it matters: large files upload quickly and the user never waits on our servers; uploading the same file twice stores nothing new.

Proof: the browser fingerprints each file while it uploads, and the server recognises an identical file: “You already uploaded this exact file” (Figure 2).

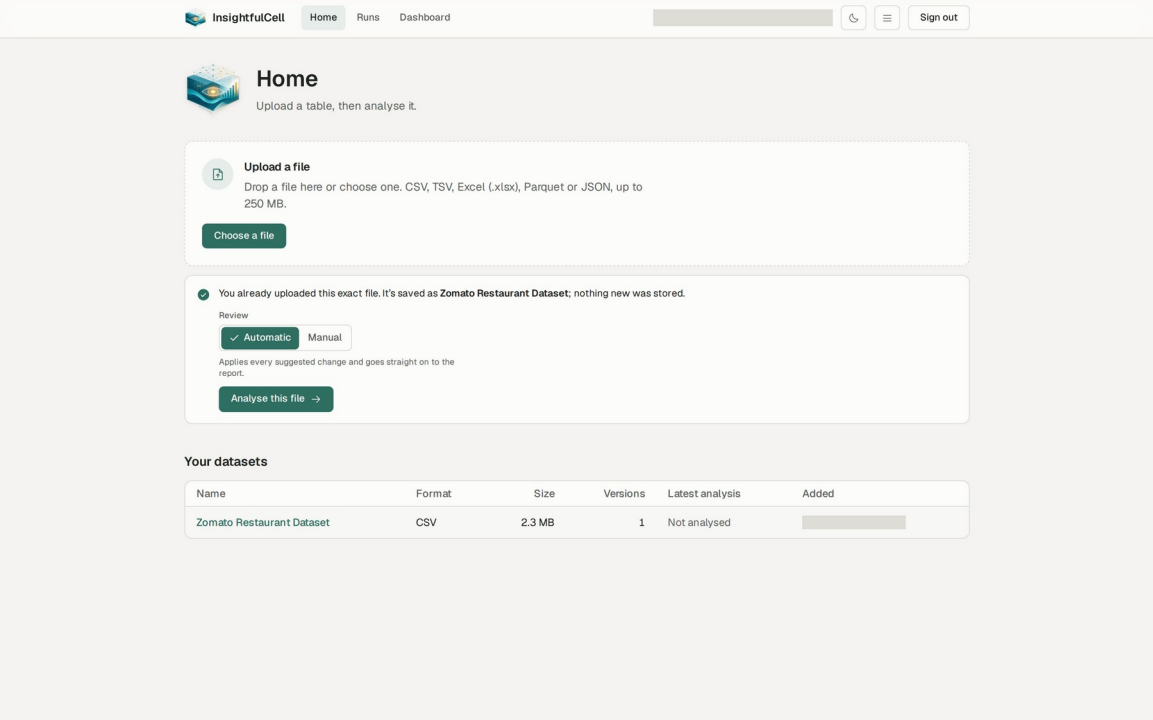


Figure 2. The Zomato file saved, and the notice shown when the same file is uploaded again

2.3 Automatic understanding and privacy protection

What it does: reads every row, decides what each column means (a measure, money, a percentage, a category, a date, a place, an identifier or free text) and masks eight kinds of personal data before anything else reads the values.

Why it matters: the type of a column decides which tests are fair, and personal data never enters the analysis.

Proof: in the case study, the restaurant names and street addresses were masked and left out of the analysis (Figure 3).

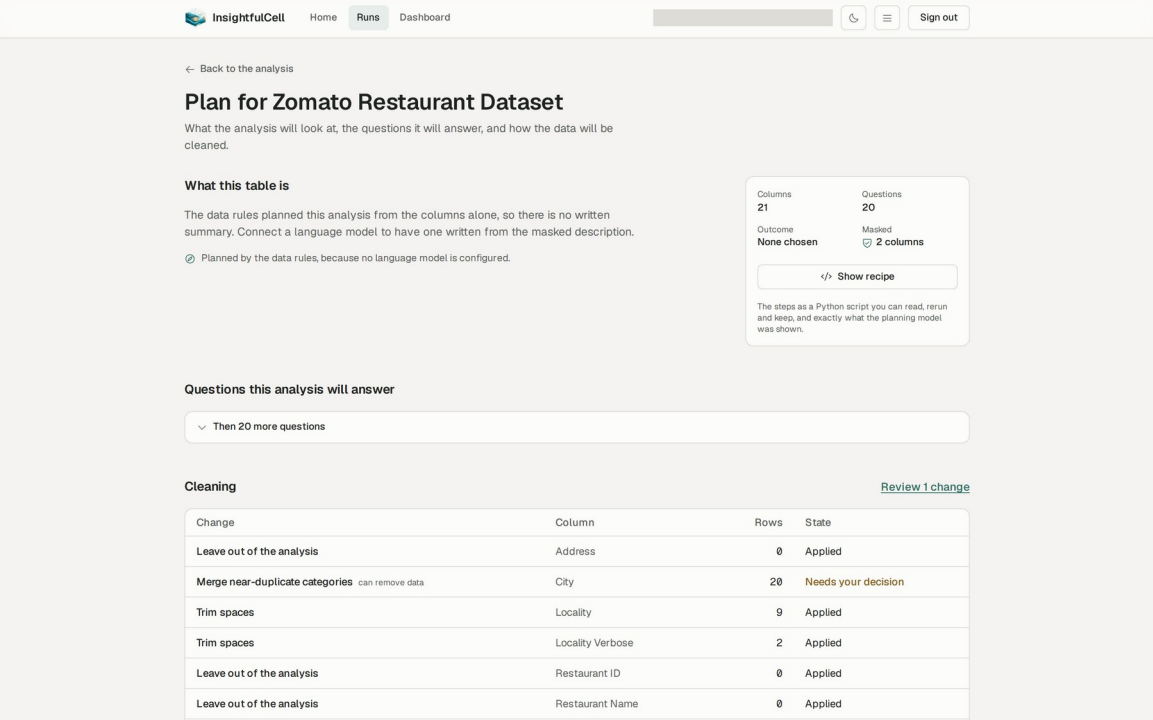


Figure 3. The analysis plan for the Zomato file, with its masked columns and the 20 planned questions

2.4 Cleaning you can review

What it does: proposes fixes of 14 kinds, each either safe (it keeps every value) or able to remove or replace values. In Automatic mode (the default) all are applied. In Manual mode the analysis stops at a review step, showing each data-changing fix with “Now” and “After” examples and the rows affected.

Why it matters: the user, not a rule, decides anything that could change the truth of the data. The original file is never modified, and a downloadable Python recipe rebuilds the cleaned table.

Proof: for the Zomato file, InsightfulCell applied 6 safe changes and stopped on one: merging “Mangalore” into “Bangalore” because the names are spelt alike (20 rows). They are different cities, so the reviewer kept the values as they were (Figure 4).

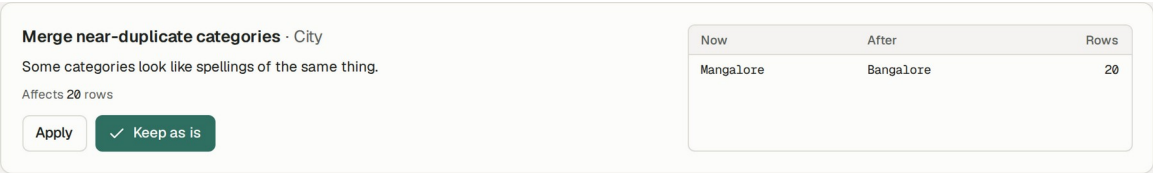


Figure 4. The review step: one proposed merge, kept as it is

2.5 Statistically tested findings

What it does: answers each planned question with one proper statistical test, using nine analysis methods (distributions, relationships, trends, group differences, unusual rows, drivers of an outcome, missing values as a signal, groups against the overall, and whether a pattern holds within groups).

Why it matters: when you run many tests, some look significant by chance; InsightfulCell limits those false alarms to 5% across the whole analysis (Benjamini & Hochberg, 1995). Each finding then gets an honest evidence tier: Strong evidence, Supported, Not reliable or Descriptive. The ten strongest reliable findings are re-tested on data with only safe changes, to show whether they depend on the cleaning.

Proof: the case study ran 10 tests and all 10 passed the control. One reads: “Price range and Aggregate rating rise together: the link is moderate (rank correlation 0.50; 95% interval 0.48 to 0.51)” (Figure 5).

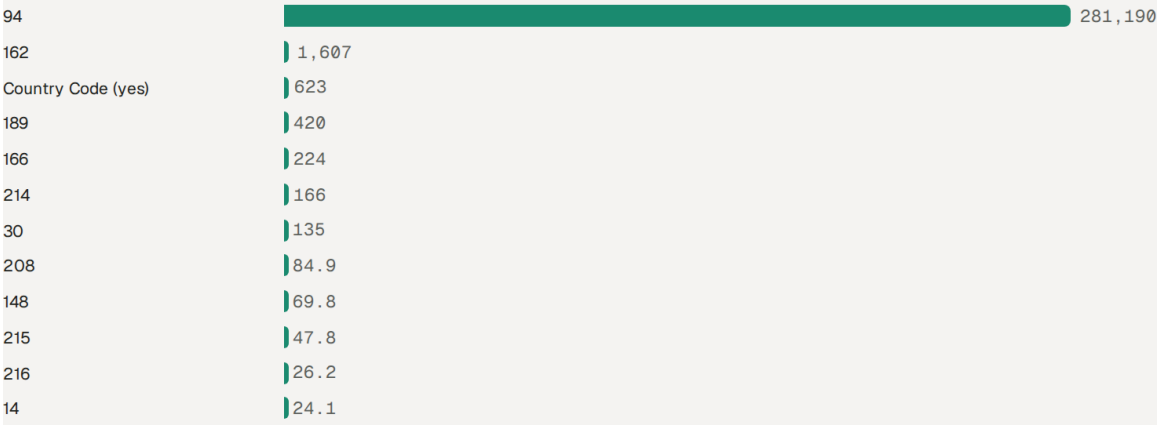
InsightfulCell Home Runs Dashboard

Comparison of groups

Does Average Cost for two differ by Country Code?

Average Average Cost for two is highest for 94 at 281,190 and lowest for 14 at 24.1, against 1,200 overall; the difference between groups is strong (eta 0.82; 95% interval 0.77 to 0.90). F1

Average Average Cost for two by Country Code



All 12 values were checked against the data Download Vega-Lite spec

Show the numbers and explore a group

Sources

F1 Average Average Cost for two is highest for 94 at 281,190 and lowest for 14 at 24.1, against 1,200 overall; the difference between groups is strong (eta 0.82; 95% interval 0.77 to 0.90). Strong evidence

Columns: Country Code, Average Cost for two 9,547 rows

Average Average Cost for two per Country Code group with at least ten rows; eta measures how much of the variation the groups account for (0 to 1).

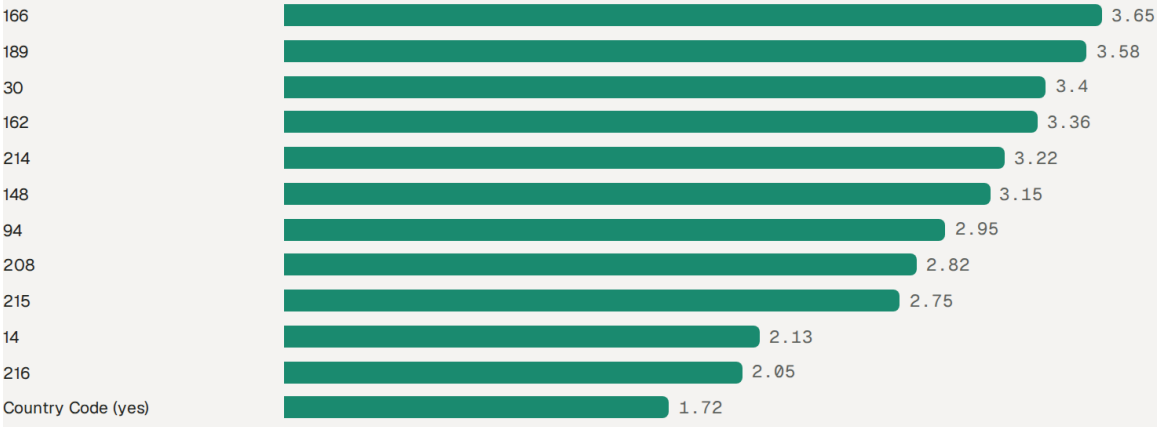
Kruskal-Wallis test; size as eta (share of variation the groups account for): Eta 0.82 (95% interval 0.77 to 0.90); p-value below 0.001 (below 0.001 after the false-discovery control).

Comparison of groups

Does Price range differ by Country Code?

Average Price range is highest for 166 at 3.65 and lowest for Country Code (yes) at 1.72, against 1.8 overall; the difference between groups is moderate (eta 0.35; 95% interval 0.33 to 0.37). F2

Average Price range by Country Code



All 12 values were checked against the data Download Vega-Lite spec

Figure 5. One answer in the report, with its checked chart and evidence badge

2.6 A report where every number shows its work

What it does: every finding is stored as an unchangeable fact with its test, rows, interval and a content hash. Sentences are written with numbered slots that the engine fills from those facts, and each sentence passes seven checks before it is kept.

Why it matters: a writer, human or AI, cannot put a wrong number into the report, and any reader can click a source chip to see the evidence.

Proof: Table 5 shows what the writer produced and what the reader sees; Figure 6 shows the fact behind it.

Table 5. The writer leaves slots; the engine fills them from the stored fact

The writer produces	The reader sees
“Price range and Aggregate rating rise together: the link is moderate (rank correlation {F10.1}; 95% interval {F10.2} to {F10.3}).”	“Price range and Aggregate rating rise together: the link is moderate (rank correlation 0.50; 95% interval 0.48 to 0.51).” F10

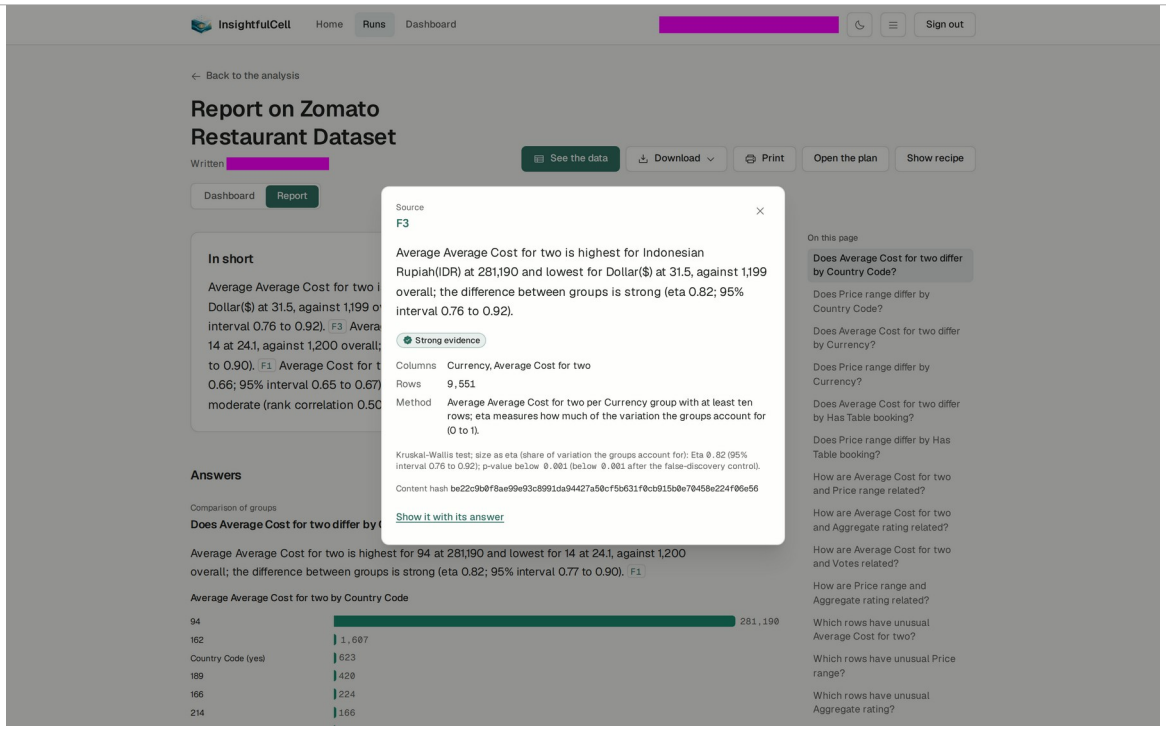


Figure 6. A source chip opened: the fact’s test, evidence tier, columns, rows and content hash

2.7 Dashboards, maps and exploration

What it does: assembles a dashboard of headline tiles, top findings and maps. Before any chart is shown, every value it draws is recomputed by its own query; a chart that disagrees is withheld. Places are matched offline to 237 countries, the states and provinces of 9 countries, and Australian postcodes. “See the data” opens the original rows beside the report, with personal columns hidden, and “Explore” runs a follow-up analysis of one group, marked exploratory.

Why it matters: no chart shows a value the data does not hold, no place name leaves the server, and users can check any finding against real rows.

Proof: the case study recomputed 1,439 values across 21 charts and 1 map and withheld none; the restaurants' coordinates matched to states of India for 85% of rows (Figure 7).

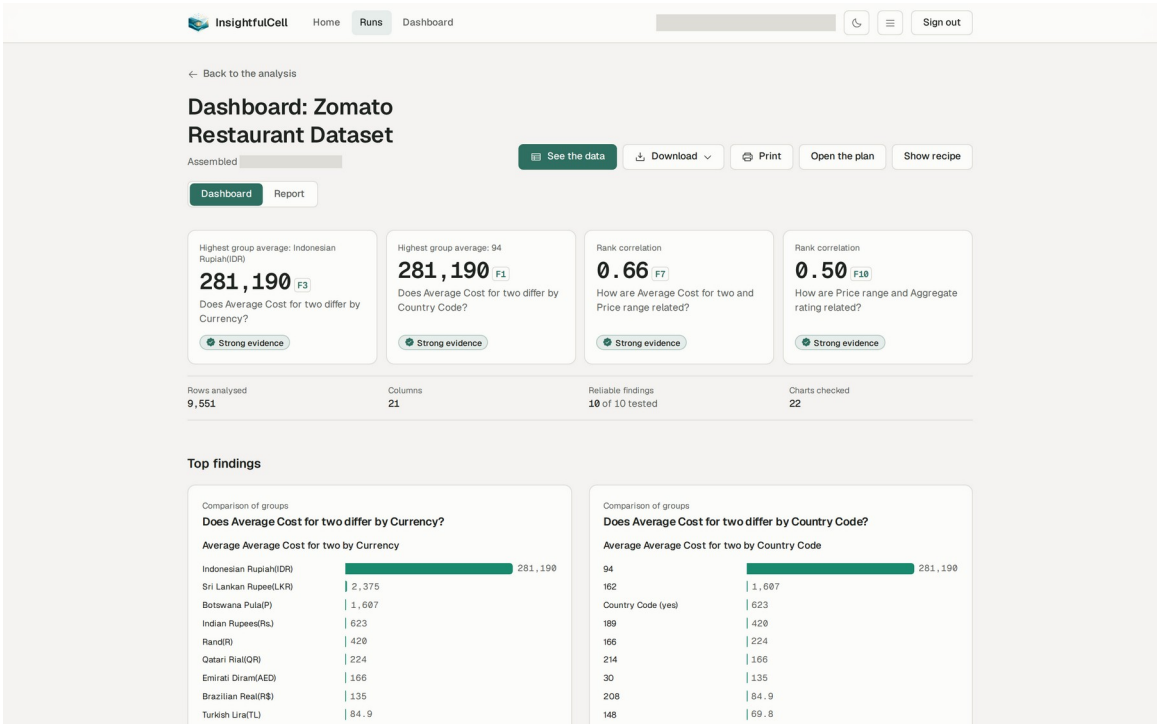


Figure 7. The Zomato dashboard: headline tiles with evidence badges and source chips

2.8 Download and share

What it does: exports the report as PDF (A4 pages), Word (editable), Markdown (plain text with tables) or HTML (one self-contained page).

Why it matters: findings travel to the people who need them in the format they already use, with every source kept.

Proof: all four formats carry the same verified sentences, charts and source list (Figure 8).

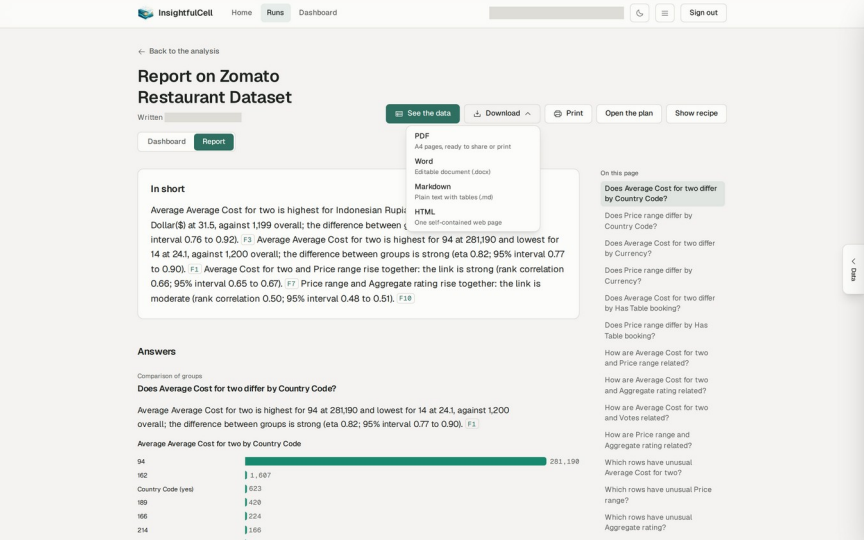


Figure 8. The Download menu with its four formats

2.9 Works with or without AI

What it does: with no AI model, built-in rules plan the analysis and the engine writes every sentence from the facts. An optional model from one of four providers can refine the plan and phrase the sentences. A backup model answers if the main provider fails, and a hard spending cap in Australian dollars is reserved before every request, so it can never be exceeded.

Why it matters: the product is resilient to provider outages and costs nothing to run without a model; adding one improves wording, not correctness.

Proof: the live site runs without any model at US$0 in model fees; with the default model, a measured analysis cost about US$0.02 (Table 6).

Table 6. Who plans and who writes in each mode

Mode	Plans the analysis	Writes the sentences	Model cost per analysis
No AI model (live site)	Built-in rules	Engine templates	US$0
Default model	Rules, refined by the model and validated	Model, verified; engine on failure	About US$0.02 (measured, 10 analyses)
Premium model	As above	As above	About US$0.26 (design budget)

2.10 Accessible by design

What it does: supports full keyboard use, screen readers, touch targets of at least 44 pixels, reduced motion, a dark theme and phone-width layouts.

Why it matters: people who use assistive technology, or only a phone, can analyse their data too.

Proof: automated accessibility checks against WCAG 2.2 level AA (World Wide Web Consortium [W3C], 2023) found 0 violations across 24 pages and states, and no page scrolls sideways at 390 pixels (Figure 9).

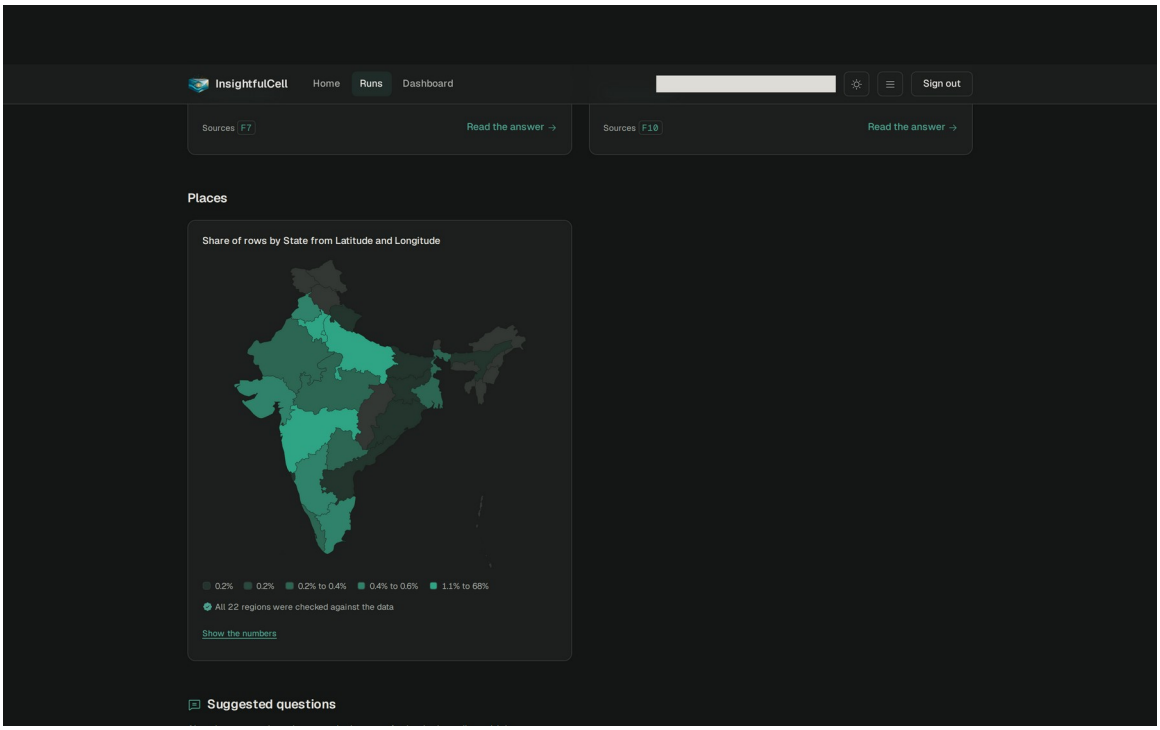


Figure 9. The dashboard in the dark theme

2.11 Handling everyday situations

Table 7 lists what users see when something unusual happens. Figure 10 shows one of them.

Table 7. Everyday situations and what the user sees

Situation	What the user sees
Wrong file type	“This file type isn’t supported”, with the accepted formats
File too large	“The file is too large. Files can be up to 250 MB.”
Duplicate upload	“You already uploaded this exact file”
Cancel and retry	“Cancel this analysis? Finished steps are kept.”; “Try again” resumes from the last saved step
Queue wait	“Queued” until the server is free (one analysis at a time on the demo)
Expired session	A return to the sign-in page; nothing is lost
Withheld chart	“A chart for this answer was withheld: one of its values did not match the data when it was checked.”

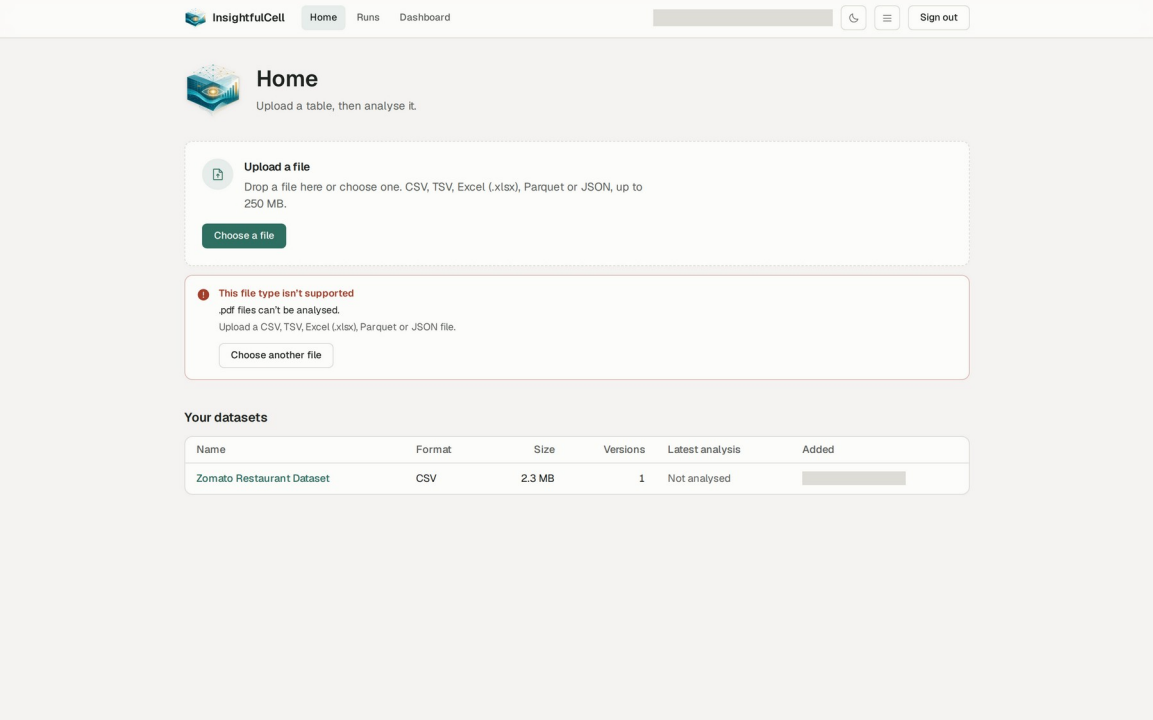


Figure 10. A PDF refused before upload

2.12 Feature summary

Table 8. Features, benefits and status

Feature	User benefit	Status
Secure upload, five formats, 250 MB, duplicate detection	Large files, no repeats	Live
Column meanings and personal-data masking	Fair tests; private data stays out	Live
Automatic and Manual cleaning, recipe	Control over every data-changing fix	Live
Nine analysis methods with false-alarm control	Findings that are not chance	Live
Verified report with source chips	Every number traceable	Live
Checked charts and offline maps	No wrong chart is ever shown	Live
See the data and Explore	Check rows; follow leads	Live
PDF, Word, Markdown and HTML	Share in any format	Live
Optional AI with backup model and spending cap	Better wording, bounded cost	Live (off on the demo)
Delete analysis, dataset or account	Users control their data	Live
Questions in plain language	Ask follow-ups in words	Next: Release 1 alpha, 11 December 2026
Team workspaces and shared reports	Work together	Next: Release 2 beta, 12 March 2027

3 Methodology: technical implementation

3.1 Development approach

We wrote the requirements and the system design before any code: what the product must do, how each component works and what happens when it fails. We then delivered in staged releases. A stage ended only when the full quality gate passed: formatting, linting, strict type checks, architecture rules, unit and integration tests on a real database, migration checks, the API contract, dependency audits and browser tests. We recorded 20 significant design decisions as decision records, each with its context, the options considered and the consequences (Nygard, 2011); three later decisions are recorded in the code and commit history.

Definition of done. A change is done when it meets its requirement, has tests for the behaviour it changes, passes every quality gate locally and in continuous integration, keeps keyboard and screen-reader behaviour intact, contains no secrets or personal data, and is documented where a user or operator needs it.

3.2 Team roles and task division

Table 9. Team roles and contributions

Member	Role	Responsibilities	Key deliverables
Rupesh Ghimire	Product Owner and Lead Engineer	Conceived InsightfulCell and sets its direction, from the problem it solves to what ships next; designed the system architecture	The web app, the API, the background worker and the analysis engine, which cleans the data, runs the statistical tests and writes the verified report; requirements, design decisions, CI and documentation
Sandesh Prasad Paudel	Quality Assurance and Deployment Engineer	Tests InsightfulCell end to end: test planning, functional and regression testing of every user flow, cross-browser, mobile, usability and accessibility checks, and acceptance testing on the live site; took it from a local setup to a live public service	AWS hosting, domain and secure HTTPS connection; the container build and deployment workflow; backups and storage access; keeping the site running for users and judges

We worked in one GitHub repository with continuous integration on every push, recorded design decisions as we made them, reviewed each change against the quality gate, and tested every release on the live site.

3.3 System architecture

InsightfulCell is one codebase that runs as three processes over two stores (Figure 11). The web app renders every page and holds the user's session; for each request to the API it mints a 15-minute signed token on the server, so the browser never holds an API credential. The API validates requests, enforces permissions and queues work. The worker runs the analysis. PostgreSQL stores the data and also runs the job queue and the live-progress events; Amazon S3 stores the files. An AI model is optional and sees only masked metadata.

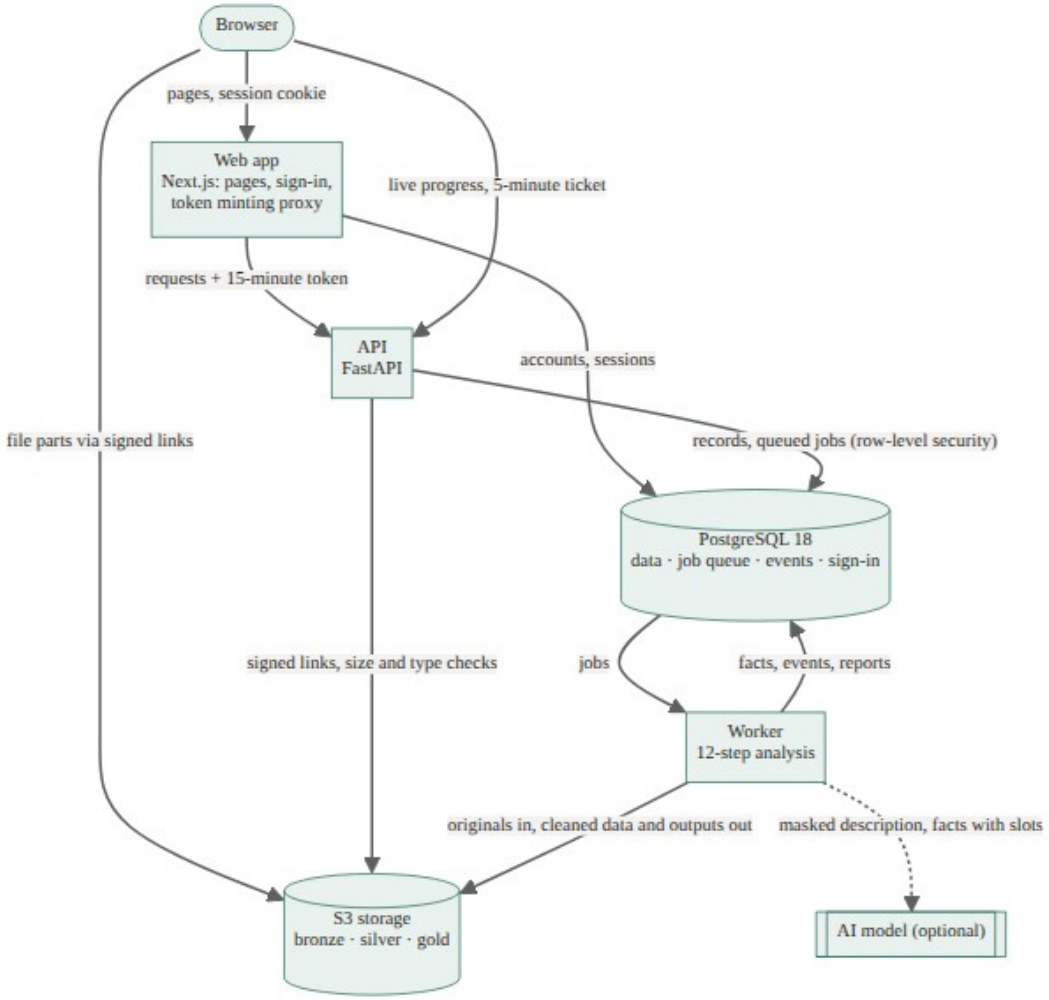


Figure 11. System architecture: one codebase, three processes, two stores

This design is simpler, safer and cheaper than a set of separate services. One package serves the API and the worker, so there is one thing to test and deploy. With no message broker or cache to run, there are fewer boundaries to secure, and isolation is enforced once, inside the database. The whole system runs on one small server and scales by adding workers.

3.4 Technology stack

Table 10. Technology choices

Layer	Technology	Why we chose it
Web app	Next.js 16, React 19, TypeScript 5, Tailwind CSS 4	Fast pages for long reports; a server-side proxy keeps credentials off the browser
Sign-in	Better Auth 1	Accounts, workspaces and signed tokens kept in our own database
API	Python 3.13, FastAPI 0.143, Pydantic 2, SQLAlchemy 2	The same language as the engine; a typed contract shared with the web app
Job queue	Procrastinate 3 on PostgreSQL	No extra system to run, secure or back up
Database	PostgreSQL 18	Row-level security, full-text search and notifications in one place
File storage	Amazon S3	Files go straight to a service built for them, with no stored keys
Analysis	DuckDB 1, NumPy 2, SciPy 1, scikit-learn 1, Shapely 2	Fast, memory-limited parsing and standard, proven statistics
Reports	Jinja2 3, python-docx 1, fpdf2 2	The same document as HTML, Word and PDF
AI (optional)	Gemini, OpenAI, Anthropic or DeepSeek behind one gateway	Any provider, or none; privacy and cost rules written once
Hosting	Docker Compose, Caddy 2, GitHub Actions	Automatic HTTPS; images built in the cloud; one-command deploys

3.5 How the analysis works

An analysis runs 12 steps in two parts, separated by the review step (Figure 12). Each step saves its result, so a retry resumes where it stopped.

1. **Understand.** Every row is read into typed columns, personal data is masked, and every column is profiled. Built-in rules plan the questions; an optional AI model may refine the plan from a masked description, and every suggestion is validated.
2. **Reviewed cleaning.** Fixes are proposed and, in Manual mode, decided by the user; they are applied to a copy, alongside a second copy with only the safe changes.
3. **Signal check.** When an outcome is chosen, a derived measure is kept only if it beats all 19 randomly shuffled copies of the outcome.
4. **Nine methods with false-alarm control.** Each question gets one primary test; all tests in the analysis are adjusted together with the Benjamini–Hochberg procedure at 5% (Benjamini & Hochberg, 1995), and every test is logged.
5. **Re-test against cleaning.** The ten strongest reliable findings are tested again on the safe-changes-only copy.
6. **Ranking.** Findings are ordered by evidence, effect size, relevance and novelty.

7. **Verified narration.** Charts are checked value by value; sentences are written with slots, filled from facts and verified, with one repair and a template fallback.

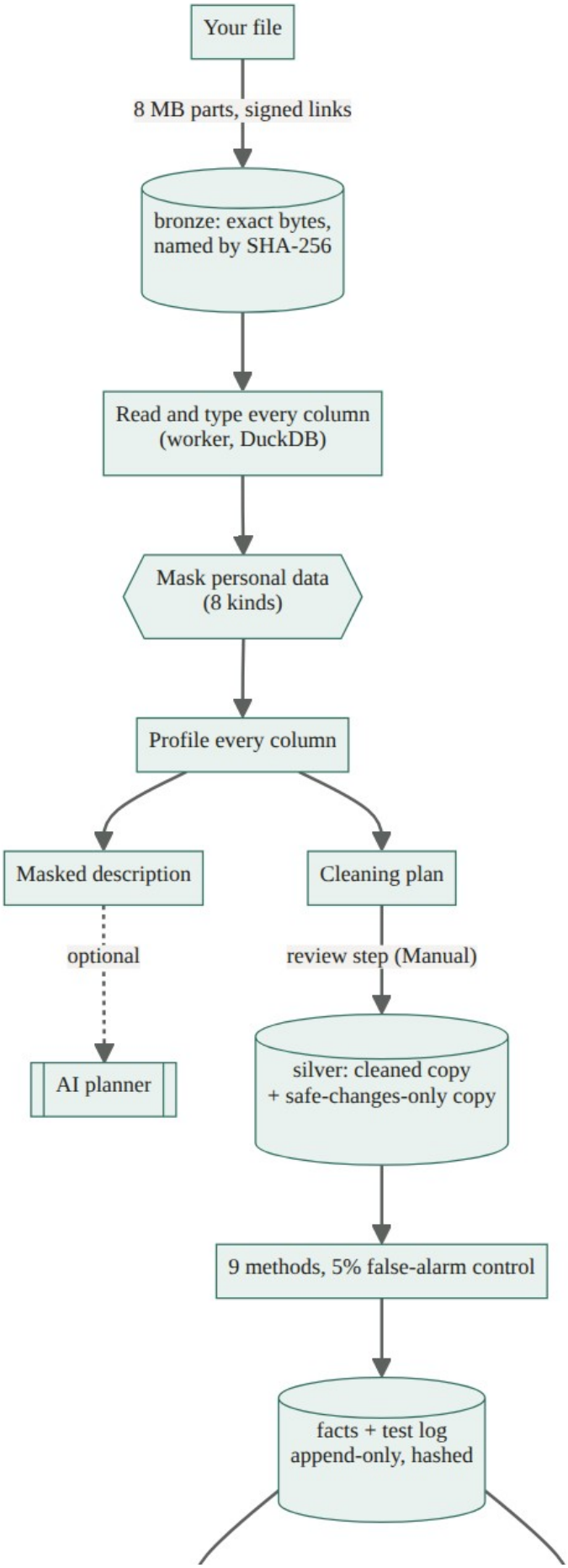

Your file
8 MB parts, signed links
bronze: exact bytes, named by SHA-256
Read and type every column (worker, DuckDB)
Mask personal data (8 kinds)
Profile every column
Masked description
Cleaning plan
optional
review step (Manual)
AI planner
silver: cleaned copy + safe-changes-only copy
9 methods, 5% false-alarm control
facts + test log append-only, hashed

Figure 12. From file to report: where data is masked, stored, transformed and checked

3.6 Software engineering principles

Table 11. Engineering principles in practice

Principle	How we applied it	Evidence
Layered design, enforced automatically	Six dependency rules checked on every change	Architecture check in the quality gate
Separation of concerns	The analysis engine contains no web, database or cloud code	The engine is tested on its own
Privacy by design	Masking at the door; no rows to AI; a final payload check	The masked description is shown to users
Least privilege	The application's database role cannot bypass row-level security; the server reaches storage through a role with no stored keys	Database role tests; deployment policy
Defence in depth	Isolation checked by the application and again by the database	0 of 22 cross-tenant attacks succeeded
Safe defaults	Works fully without AI; the AI is optional and capped	The live site runs with no model
Records that cannot be altered	Facts, tests, events and reports are append-only with content hashes	Database triggers reject changes
Resumable jobs	12 saved steps; retries skip finished work	Worker crash tests
Automated testing and CI/CD	1,275 automated tests; images built on every change	Quality gate results (Table 13)
Reproducible deployment	Images tagged by commit; one-command deploy and rollback	Deployment scripts
Accessibility by design	Keyboard, labels, contrast and motion settings, checked automatically	0 violations on 24 pages
Documented decisions	20 decision records with options and consequences	Decision records

3.7 Security and privacy by design

Table 12 maps the main risks to their controls. The controls address several OWASP Top 10 categories (OWASP Foundation, 2021), including broken access control, injection and security misconfiguration. Collecting only what is needed, masking personal data and deleting on request follow the intent of the Australian Privacy Principles (Office of the Australian Information Commissioner [OAIC], n.d.); InsightfulCell has not yet had a formal privacy assessment.

Table 12. Risks, controls and evidence

Risk	Our control	Evidence
One customer seeing another's data	The workspace comes only from the verified token; forced row-level security on every table (PostgreSQL Global Development Group, n.d.)	0 of 22 cross-tenant attacks succeeded

Risk	Our control	Evidence
Stolen login tokens	API tokens never reach the browser and expire in 15 minutes; cookies cannot be read by scripts; sign-out ends access at once	Replayed cookies are refused after sign-out
Personal data reaching an AI model	Masked at ingest; only a masked description is shared; requests carrying a personal value are blocked	Masked description shown in the app
Malicious files	Type allow-list, content check, size, row and column limits, archive guard, parsing only in the worker	Disguised and oversized files refused
Tampering with results	Append-only records with content hashes	Hash shown in every fact window
Data leaking in errors or logs	Written messages never echo cell values; logs carry no prompts or values; standard problem responses (Nottingham et al., 2023)	A row holding a secret produced only a line number
Runaway AI cost	The worst-case cost of every request is reserved against hard caps first	Spending ledger tests
Admin access	A separate long key, asked on every visit, compared in constant time and rate-limited	Browser tests of the admin page

3.8 Testing and quality assurance

Testing runs at three levels (Figure 13): 726 backend tests on a real database and storage service, 537 frontend component tests (with automated accessibility checks), and 12 browser tests that drive a complete fresh stack. Specialised evaluations run with every test suite: planted-effect datasets that the statistics must find, and a corrupted narrator whose 31 planted errors the verifier must reject.

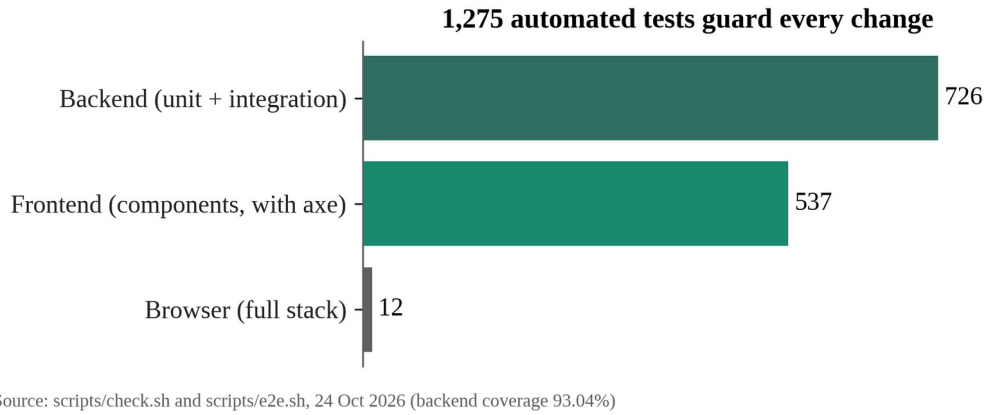


Figure 13. Automated tests by suite

Table 13. Quality gates and latest results

Quality gate	Tool	Latest result
Style and lint	ruff, Prettier, ESLint with accessibility rules	Pass (24 Oct 2026)
Types	mypy (strict), TypeScript (strict)	Pass (24 Oct 2026)
Architecture	import-linter, 6 rules	Pass (24 Oct 2026)
Backend tests	pytest on PostgreSQL 18 and S3-compatible storage	726 passed, 93.04% coverage (24 Oct 2026)
Frontend tests	Vitest, Testing Library, axe	537 tests; 536 passed, 1 awaiting a name update (24 Oct 2026)
Browser tests	Playwright on a fresh full stack	12 of 12 passed (24 Oct 2026)
Accessibility tour	axe, WCAG 2.2 AA, 24 pages	0 violations (24 Oct 2026)
Supply chain	pip-audit, pnpm audit, secret scan	No known vulnerabilities; no secrets (24 Oct 2026)
Security battery	99 live attacks with two to four accounts	92 passed; 0 of 22 cross-tenant attacks succeeded (18 Oct 2026)
Live check	Full user flow on the public site	Every step passed (22 Oct 2026)

3.9 Code quality and documentation

The repository holds the web app, the backend (API, worker and analysis engine), the deployment kit, scripts and sample data. All code is typed and linted, and the six architecture rules stop one layer from reaching into another. The README covers installation, architecture, tests, deployment, design decisions, team contributions and licence; a usage guide explains every screen and the logic behind it. A judge can run InsightfulCell locally with five commands (Appendix G).

3.10 Deployment and operations

The live site runs on one Amazon EC2 t3.micro server under Docker Compose (Figure 14). Caddy obtains and renews the HTTPS certificate automatically. Uploaded files live in a private, encrypted Amazon S3 bucket reached through the server’s role, so no storage keys are stored anywhere. GitHub Actions builds the container images on every change; the server pulls them, and one command deploys a release or rolls back to an exact earlier build. A nightly job backs the database up to a second S3 bucket. The server and its disk sit inside the AWS free tier; the public IP address costs about US$3.60 a month (estimated).

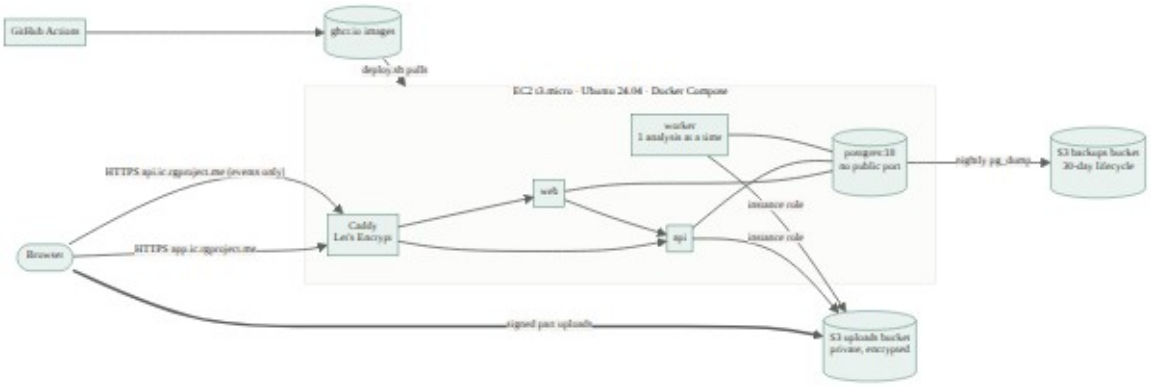


Figure 14. Deployment on AWS

4 Discussion

4.1 Market need and opportunity

The 2,814,778 actively trading Australian businesses, 996,203 of them employers (ABS, 2026), are the clearest sign of the need: almost none can justify a full-time analyst, yet all of them make decisions from data. The same holds for community organisations, schools and public-sector teams. At the same time, the unreliability of AI-generated text is well documented (Ji et al., 2023), and Australian privacy law expects organisations to protect personal information they hold (OAIC, n.d.). A tool that is easy, private and checkable meets all three pressures at once.

4.2 Target customers and use cases

Table 14. Target segments

Segment	Their pain	What InsightfulCell gives them	Example use
Small and medium businesses	No analyst; no time	A checked report from a spreadsheet in minutes	Which products and stores drive margin
NGOs and public sector	Sensitive data; strict accountability	Private analysis with a full audit trail	Program outcomes by region
Students and researchers	Statistics are easy to misuse	Correct tests, honest tiers, a reproducible recipe	A first analysis of survey data
Consultants and accountants	Clients need defensible numbers	Reports where every number shows its source	Client-ready findings in Word or PDF

4.3 Competitive positioning

Table 15 compares InsightfulCell with the two common categories of tool. Only InsightfulCell meets all six needs.

Table 15. Competitive positioning

Need	InsightfulCell	General AI assistants	BI and statistics tools
Every number traceable	Yes	No	Partly
Raw rows stay private	Yes	No	Yes
No analyst needed	Yes	Yes	No
False discoveries controlled	Yes	No	Only if the analyst does it
Reproducible cleaning recipe	Yes	No	Partly
Plain-language report	Yes	Yes	No

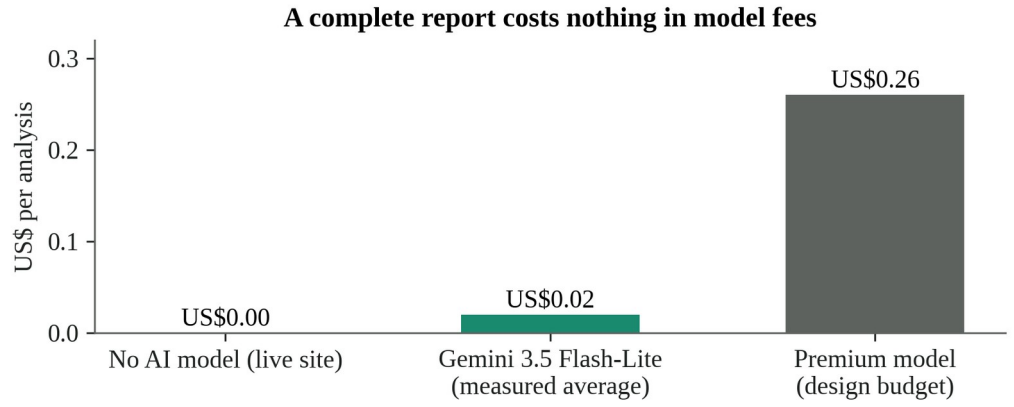
4.4 Business model and cost to serve

Proposed; pricing not yet validated. Table 16 sets out three proposed tiers. Because the rules-only mode is complete and costs nothing in model fees, a free tier is sustainable; AI planning and writing become a paid improvement with a per-person spending cap.

Table 16. Proposed tiers and cost to serve

Tier (proposed)	Offer	Model cost per analysis	Hosting
Community	Free; rules plan and the engine writes	US$0	Shared
Pro	Per seat; AI-planned and AI-written, per-person cap	About US$0.02 (default model) to about US$0.26 (premium model)	Shared
Team / private	Per organisation; team workspaces, roles, self-hosting	As Pro, or the customer's own model	Dedicated or self-hosted

Today's hosting costs about US$3.60 a month (estimated). Figure 15 shows the model cost per analysis in each mode.



Source: Live configuration; local spend ledger (10 analyses, 19 and 24 Oct 2026); system design v1.1

Figure 15. Model cost per analysis by mode

4.5 Scalability and go-to-market

The system design plans for 200 organisations, 2,000 active users and 20,000 analyses a month in the first year, with 99.5% API availability. The path there needs no redesign: a larger server and more worker processes first, then workers on their own servers scaling with the length of the queue, then a managed database and container services. Workers hold no state, so adding them is a configuration change.

We propose a staged launch: a student and university pilot (where the honest statistics teach good practice), then consultants and accountants (who need defensible numbers for clients), then small businesses and community organisations.

4.6 Ethical and social impact

- **Privacy:** personal data is masked at the door and never analysed; raw rows never reach an AI model; users can delete everything at any time.
- **Responsible AI:** an AI model cannot write a number, causal claims are rejected, and evidence tiers say plainly when a result is not reliable.
- **Statistical integrity:** false alarms are controlled across every analysis, small groups are never reported alone, and findings that depend on cleaning are flagged.
- **Accessibility:** the product works by keyboard, with screen readers and on phones.
- **Demo data:** the samples are public-domain datasets and synthetic test files, with no private individuals.
- **Opening up analysis:** organisations without analysts can make decisions from tested evidence instead of guesses.

4.7 Current limitations

Table 17. Current limitations and planned fixes

Limitation	Planned fix	When
Automatic cleaning can mis-read a few rare formats: percentages stored as 0–100, decimal commas, and similar-looking names (for example “Austria” and “Australia”). Manual review lets you approve every change.	Better detection rules; changes that can lose data always shown	Before Release 1 alpha (11 Dec 2026)
A few file shapes cannot be read yet: nested JSON values, Excel sheets with title rows above the header, quoted headers containing line breaks.	Wider file support	Before Release 1 alpha
One small server runs one analysis at a time, with no staging, monitoring or audit log yet.	A larger server or second worker; staging, monitoring, audit log, rate limits	Release 1 alpha (11 Dec 2026)
One workspace per person, with no invitations or shared reports.	Team workspaces and roles	Release 2 beta (12 Mar 2027)
You cannot yet ask questions of your data in plain language.	Questions about your data, verified like the report	Release 1 alpha (11 Dec 2026)

4.8 Future improvements and roadmap

Table 18. Roadmap

Release	Target date	What it delivers to users
Release 1 alpha	11 December 2026	Plain-language questions about your data; safer automatic cleaning; staging, monitoring, audit log and rate limits

Release	Target date	What it delivers to users
Release 2 beta	12 March 2027	Team workspaces with roles and invitations; shared reports; search by meaning; load and penetration testing
General availability	9 April 2027	Service targets of 99.5% API availability and 98% successful analyses; compliance documentation

4.9 Risks and mitigations

Table 19. Top risks

Risk	Likelihood	Impact	Mitigation
Automatic cleaning changes data unnoticed	Medium	High	Manual review today; safer rules and always-shown data-changing fixes before Release 1
The single server fails or is overloaded	Medium	Medium	Nightly backups, one-command rebuild; separate workers and a managed database at scale
AI providers change models or prices	High	Low	Four providers, a backup model, a hard cap, and a complete product without AI
Slow adoption	Medium	High	A free tier; a university pilot; reports in formats people already use
Hosting cost grows with use	Low	Medium	Stateless workers scale with demand; rules-only analyses cost nothing in model fees

4.10 SWOT summary

Table 20. SWOT summary

Strengths	Weaknesses
Every number traceable to a test; verified sentences	Single server today
Privacy by design; works fully without AI	Rare cleaning formats need Manual review
Statistical control across each analysis	No team features yet
Live, tested, low-cost deployment	Brand new; no paying users
Opportunities	**Threats**
2.8 million Australian businesses with data but no analyst	AI assistants adding analysis features
Growing concern about AI accuracy and privacy	Large BI vendors simplifying their tools
Education and research pilots	Changes in AI model pricing and terms

5 Conclusion

People with data face a trust gap: easy AI tools can invent numbers and need raw data, while accurate tools need an analyst. InsightfulCell closes that gap. Code computes every number, an AI model may only plan and phrase, and every sentence and chart is checked against stored facts before anyone sees it. Personal data is masked at the door, raw rows never reach an AI model, and the product works fully without one.

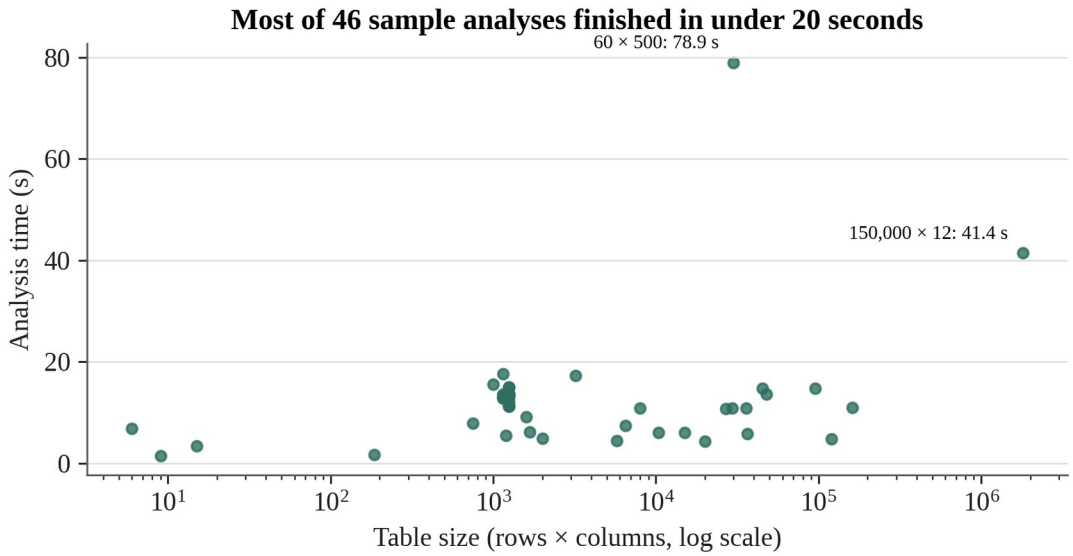
The proof is measurable: a 5,000-row file goes from upload to report in 14 seconds on a free-tier server; 0 of 22 cross-tenant attacks succeeded; the verifier rejected all 31 planted errors; the Zomato case study recomputed 1,439 chart values and withheld none; and 1,275 automated tests guard every change. It costs nothing in model fees to run, and about US$3.60 a month to host.

Next, users will ask questions of their data in plain language (Release 1 alpha, 11 December 2026), then work together in team workspaces (Release 2 beta, 12 March 2027).

InsightfulCell is a one-stop solution for data analysis. Bring a spreadsheet; leave with clear answers, charts and a report your team can check.

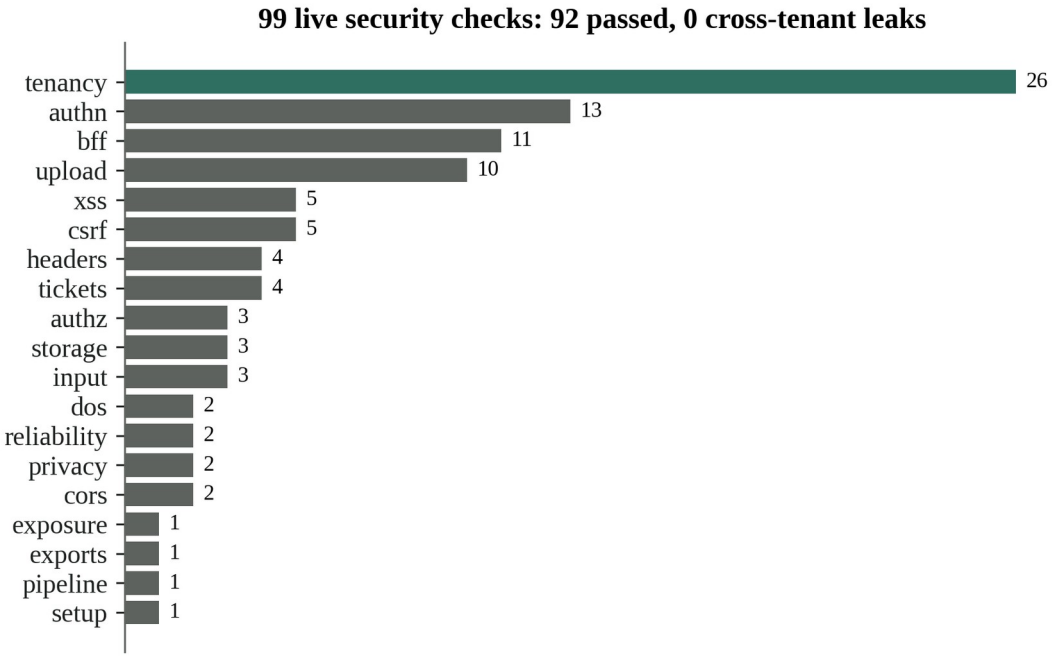
Appendices

Appendix A. Evidence charts



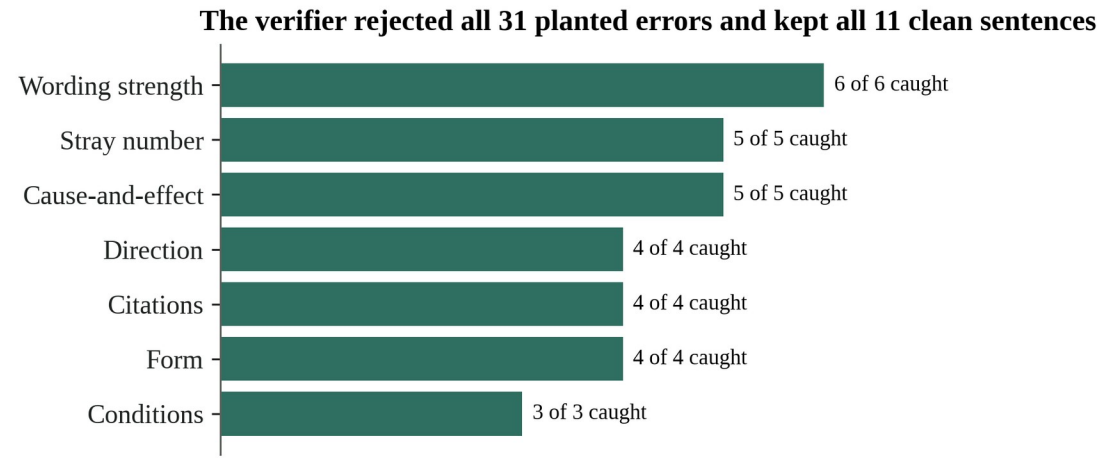
Source: InsightfulCell system test, 18 Oct 2026 (worker time, rules planner, local machine)

Figure A1. Analysis time against table size across 46 sample analyses



Source: InsightfulCell system test, 18 Oct 2026 (two to four attacker accounts)

Figure A2. Live security checks by category



Source: backend/tests/unit/test_narration.py (corrupted-narrator evaluation), run 24 Oct 2026

Figure A3. Verifier evaluation by check

Appendix B. Delivery timeline

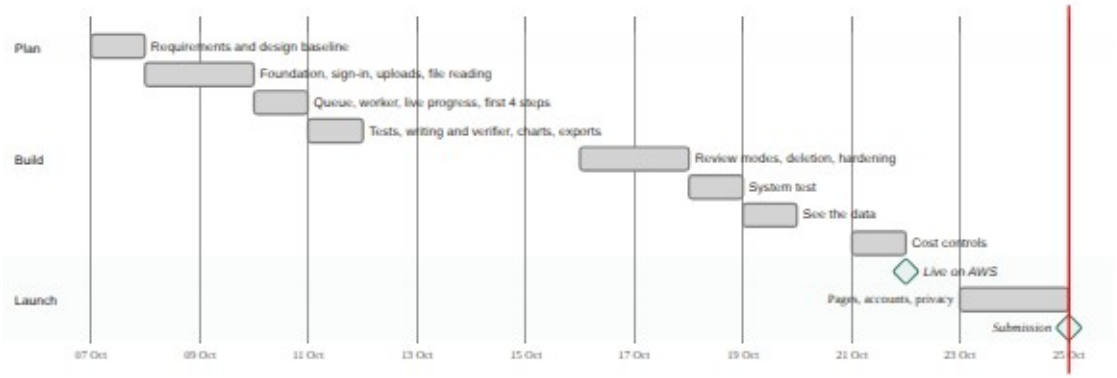


Figure B1. Delivery timeline, 7–25 October 2026

Appendix C. Additional screenshots

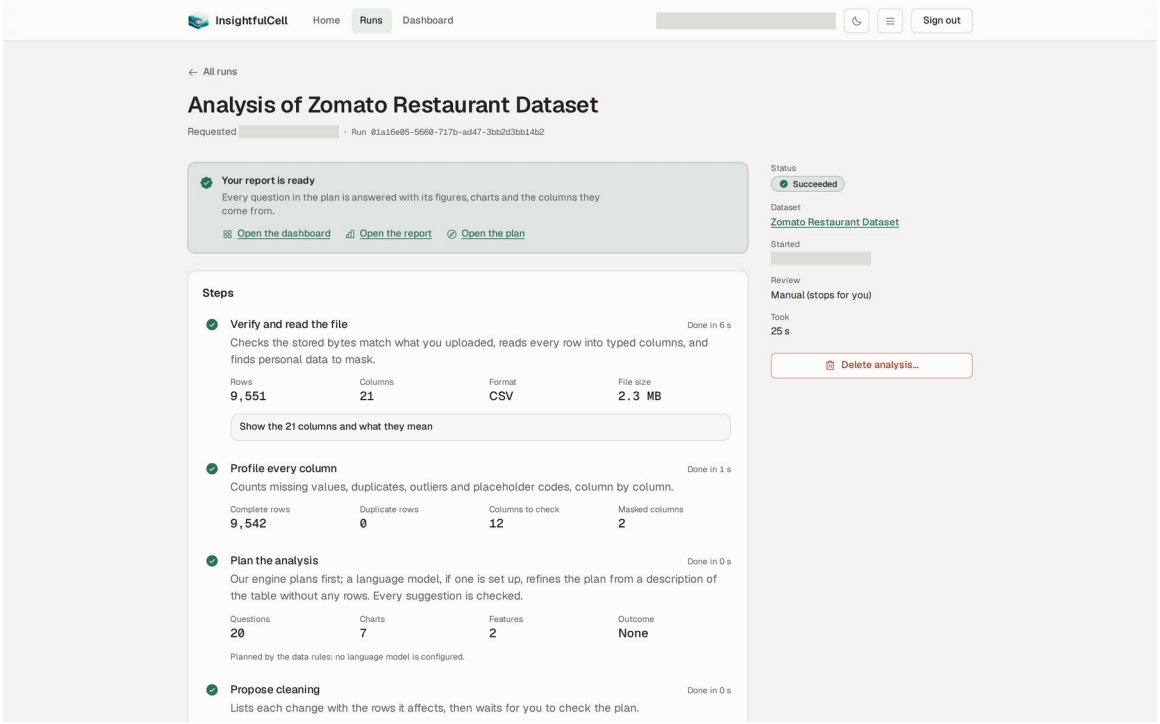


Figure C1. Live progress through the 12 steps

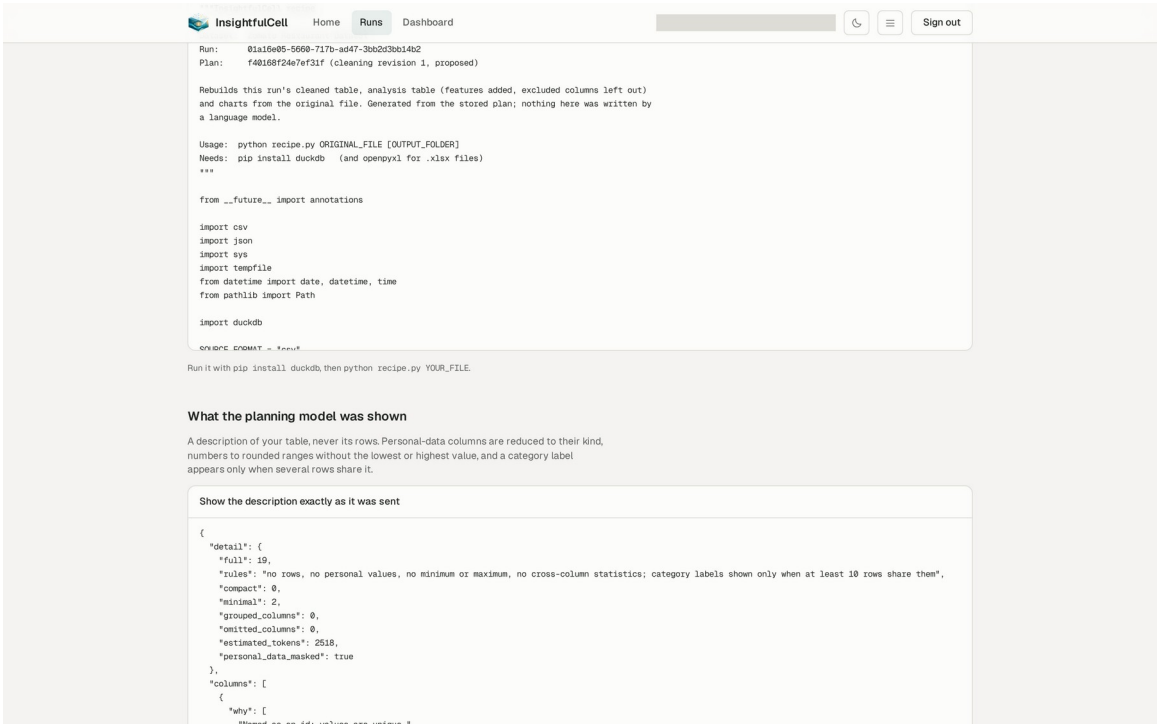


Figure C2. The recipe and the masked description a model would be shown

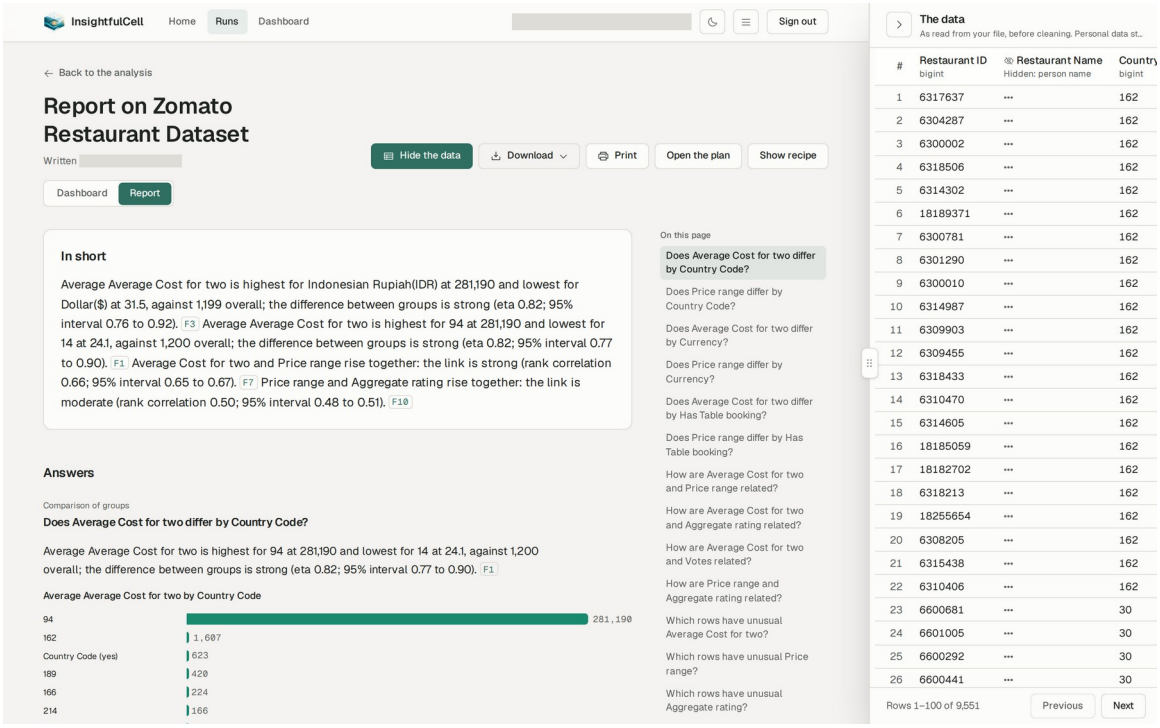


Figure C3. See the data beside the report, personal columns hidden

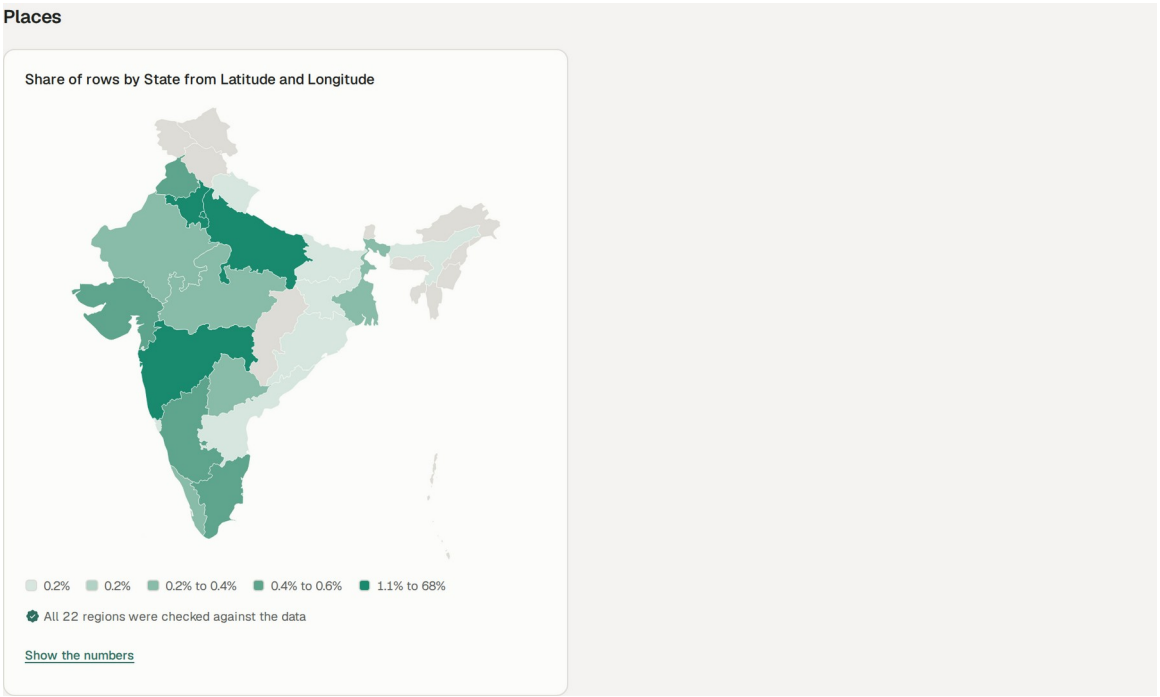


Figure C4. The map drawn from the restaurants' coordinates

Appendix D. Sample output

The report's opening for the Zomato case study ("In short"), as written by the engine on 25 October 2026:

> Average Cost for two and Price range rise together: the link is strong (rank correlation 0.66; 95% interval 0.65 to 0.67). F7

Price range and Aggregate rating rise together: the link is moderate (rank correlation 0.50; 95% interval 0.48 to 0.51). F10

The same report also found that “Average Cost for two” differs strongly by currency (eta 0.82). This is correct but expected, because the column mixes 12 currencies; it is the kind of comparison a reviewer can leave out in Manual mode.

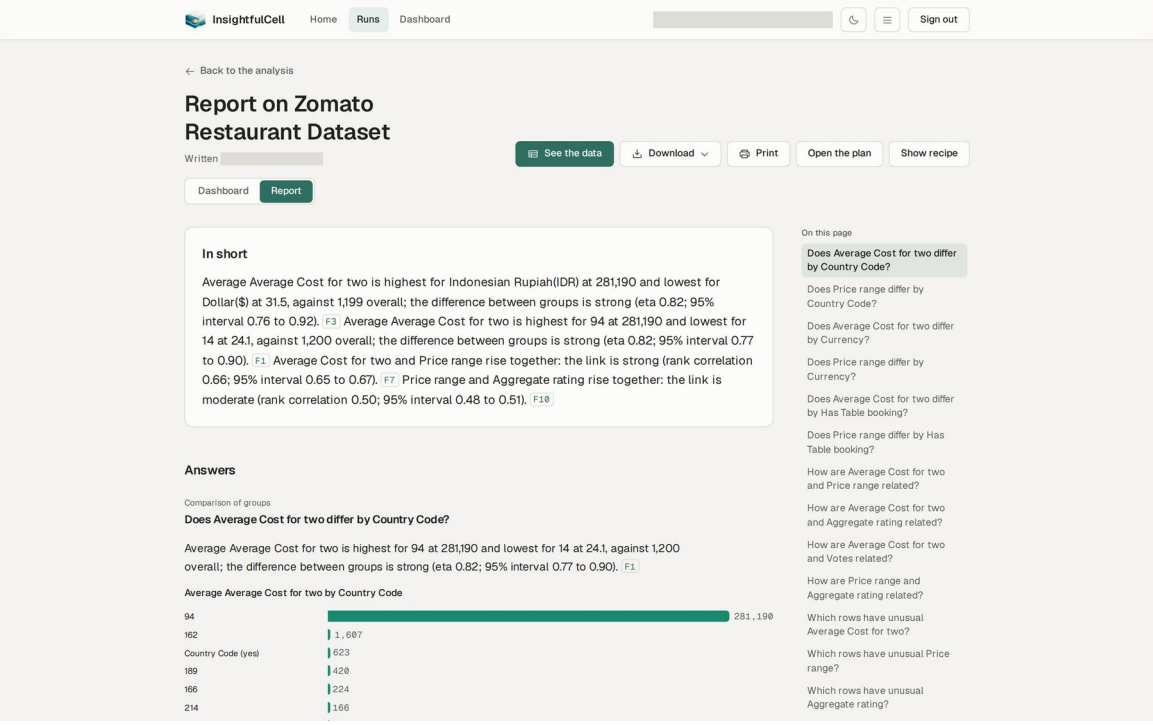


Figure D1. The report’s first page

Appendix E. Technology detail

Table E1. Full versions (24 October 2026)

Component	Version
Next.js / React / TypeScript / Tailwind CSS	16.4.3 / 19.3.0 / 5.9 / 4.3.3
Better Auth	1.7.7
Python / FastAPI / Pydantic / SQLAlchemy	3.13 / 0.143.0 / 2.14.0 / 2.1.4
Procrastinate / PostgreSQL	3.10.0 / 18
DuckDB / NumPy / SciPy / scikit-learn / Shapely	1.5.6 / 2.5.3 / 1.18.1 / 1.9.1 / 2.2.0
fpdf2 / python-docx / Jinja2	2.8.9 / 1.2.0 / 3.1.6
pytest / Vitest / Playwright	9 / 5.0.3 / 1.56.1
Caddy / Docker Compose	2 / v2

Appendix F. Glossary

Table F1. Glossary

Term	Meaning
Benjamini–Hochberg	A method that adjusts many test results together to limit the share of false alarms

Term	Meaning
Content hash	A fingerprint of a record that changes if any part of it changes
Effect size	How large a difference or link is, separate from whether it is significant
Evidence tier	Strong evidence, Supported, Not reliable or Descriptive
Exploration	A follow-up analysis of one group, marked exploratory
Fact	A stored, unchangeable record of one finding with its test and source rows
False alarm	A result that looks significant only by chance
Masked	Hidden from analysis and from any AI model because it is personal data
Recipe	A Python script that rebuilds the cleaned table
Review step	The pause in Manual mode where the user decides each data-changing fix
Row-level security	A database rule that stops one customer reading another’s data
Slot	A placeholder in a sentence that code fills with a number from a fact
Source chip	The clickable fact id beside a number in the report
Workspace	A user’s private space for files and analyses

Appendix G. How to evaluate InsightfulCell

1. Open the live application: https://app.ic.rgproject.me.
2. Create an account with any email address you can receive mail at, and choose a password.
3. Download the sample file Zomato Restaurant Dataset.csv; other samples are in the repository’s sample data folder.
4. Follow the usage guide, which explains every screen and the logic behind it.
5. To run it locally, follow the README’s installation section: install the prerequisites, run `make bootstrap`, `make up`, `make migrate` and `make dev`, then open http://localhost:3000.

Appendix H. Acknowledgements and attribution

Table H1. Key open-source libraries

Library	Used for	Licence
Next.js, React, Better Auth, Tailwind CSS	Web app and sign-in	MIT
FastAPI, Pydantic, SQLAlchemy, Procrastinate, DuckDB	API, data access, job queue, parsing	MIT
psycopg, fpdf2	Database driver, PDF reports	LGPL-3.0

Library	Used for	Licence
NumPy, SciPy, scikit-learn, Shapely, Jinja2	Statistics, geometry, templates	BSD-3-Clause
boto3, Google Gen AI SDK, OpenAI SDK, Playwright	Storage, optional AI, browser tests	Apache-2.0
axe-core	Accessibility tests	MPL-2.0

- **Reference data:** country and state boundaries from Natural Earth 5.1.2 (public domain).
- **Fonts:** DejaVu Sans in exported reports (DejaVu fonts licence) and Geist in the web app (SIL Open Font Licence).
- **Sample data:** the Zomato Restaurants Data, AI Usage and Impact on Students & Professionals, and Data Science Job Listings – Australia datasets are public-domain (CC0) datasets from Kaggle. All other sample files are synthetic, generated for testing.
- **AI-assisted development:** we used AI coding assistants during development; our own contributions and design decisions are set out in the repository's README, under "Team and contributions".

References

Australian Bureau of Statistics. (2026, August 18). *Counts of Australian businesses, including entries and exits, July 2022 – June 2026*. https://www.abs.gov.au/statistics/economy/business-indicators/counts-australian-businesses-including-entries-and-exits/latest-release

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological), 57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys, 55*(12), 1–38. https://doi.org/10.1145/3571730

Nottingham, M., Wilde, E., & Dalal, S. (2023). *Problem details for HTTP APIs* (RFC 9457). Internet Engineering Task Force. https://www.rfc-editor.org/rfc/rfc9457

Nygard, M. (2011, November 15). *Documenting architecture decisions*. Cognitect. https://cognitect.com/blog/2011/11/15/documenting-architecture-decisions

Office of the Australian Information Commissioner. (n.d.). *Australian Privacy Principles*. Retrieved 25 October 2026, from https://www.oaic.gov.au/privacy/australian-privacy-principles

OWASP Foundation. (2021). *OWASP Top 10: 2021*. https://owasp.org/Top10/

PostgreSQL Global Development Group. (n.d.). *Row security policies*. PostgreSQL documentation. Retrieved 25 October 2026, from https://www.postgresql.org/docs/current/ddl-rowsecurity.html

World Wide Web Consortium. (2023). *Web Content Accessibility Guidelines (WCAG) 2.2*. https://www.w3.org/TR/WCAG22/